

ZEROPAY DIGITAL SERVICES LTD

United Kingdom Subsidiary of BETONIQ WEST
LTD (RC 1496603, Nigeria)

New Ventures Technical White Paper

*AgriLink · CivicTrack · TriageAI · LearnOffline ·
SafeGuard · ChipLink*

**Prepared by: Tom Ugbodaga — Shareholder &
CEO**

MSc Computer Systems Security (University of
Gloucestershire — England, 2011)

LinkedIn: [Tom to confirm profile URL for insertion]

September 2026

*Prepared for grant, government, and institutional
review — Tech Grants / Government Grants / EU
Grants*

Executive Summary

Zeropay Digital Services Ltd is a United Kingdom subsidiary of BETONIQ WEST LTD, established to develop and operate six independent software solutions, each targeting a distinct, urgent global problem: food loss in agriculture, corruption and delay in public services, early cancer/disease detection access, offline education access, fraud/trafficking prevention, and semiconductor supply-chain concentration. All six platforms are engineered independently of any single vendor platform, using GitHub for version control and Hugging Face for AI-model hosting and inference, ensuring long-term technical sovereignty and portability.

Each platform below is presented with its problem statement, technical architecture, core data model, security and privacy posture, current build status, and its fit against active technology, government, and EU grant funding lines.

1. AgriLink

Real-time farmer-to-buyer and cold-chain logistics marketplace.

Key statistic: 1.3 billion tonnes of food — roughly one third of everything the world produces — is lost or wasted every year, while 733 million people go hungry (FAO/UNEP data).

Problem Statement

Smallholder farmers in emerging markets routinely lose 30-40% of harvested crops before reaching a buyer, due to the absence of real-time market information, unreliable transport coordination, and multiple unnecessary broker layers that both delay sale and depress farmer income. AgriLink removes the information and logistics gap directly.

Technical Architecture

- Geolocation-ranked matching engine surfaces nearby listings to buyers and available trucks in real time.
- Lightweight mobile-first client designed for low-end Android devices and intermittent connectivity.
- Freshness-window scoring de-prioritizes listings approaching spoilage risk to encourage fast matching.

- In-app deal confirmation replaces informal broker negotiation with a timestamped digital record.

Core Data Model

Entity	Purpose
Farmer	Profile, location, active crop listings, verified transaction history.
CropListing	Crop type, quantity, harvest date, asking price, freshness window.
Buyer	Verified buyer/aggregator profile and purchase history.
LogisticsTruck	Real-time location, capacity, cold-chain capability, availability.
Transaction	Confirmed deal record: price, quantity, pickup time, settlement status.

Security, Privacy & Compliance

- Farmer and buyer personal data minimized to what is operationally necessary (name, phone, location).

- Transaction history immutable once confirmed, to support dispute resolution and lender credit assessments.
- Role-based access ensures buyers cannot see other buyers' bid data (sealed-bid style fairness).

Feasibility & Build Status

MVP data model and matching-engine logic specified. Frontend and API build proceeding on GitHub under Zeropay Digital Services Ltd; AI-assisted matching/ranking model to be hosted on Hugging Face once deployment token is connected.

Grant & Funder Fit

Strong fit for agri-tech and food-security grant lines: IFAD Agri-Tech innovation funds, GSMA AgriTech Innovation Fund, EU Horizon Europe Food Systems calls, and national agriculture ministry digitization grants.

2. CivicTrack

Transparent digital tracking for public service applications.

Key statistic: Corruption is estimated to drain over \$3.6 trillion from the global economy every year (UNODC), much of it extracted through delay and informal 'facilitation fees' in basic public services.

Problem Statement

Citizens applying for permits, identity documents, and licenses frequently face undocumented delay, creating the exact conditions in which informal payment is solicited to 'speed up' a process. Without a visible, timestamped status trail, there is no accountability mechanism for either the citizen or the reviewing officer.

Technical Architecture

- Every submission generates an immutable, timestamped reference number visible to the applicant instantly.
- Agency-side dashboard restricts status changes to authenticated staff, each change hash-chained to the prior state for tamper evidence.
- Aggregate average-processing-time analytics published per service type and per agency, creating public accountability pressure.

- SMS/USSD status-check fallback for applicants without smartphone access.

Core Data Model

Entity	Purpose
Applicant	Citizen identity, submitted requests, contact channel.
ServiceRequest	Request type, reference number, current status, submission date.
Agency	Government department/office handling a request type.
StatusUpdate	Timestamped state change, responsible officer, optional note.

Security, Privacy & Compliance

- Hash-chained status history makes retroactive tampering with the audit trail detectable.
- Officer-level access logging tied to individual staff accounts, not shared logins.
- Citizen data handled under data-minimization principles appropriate to public-sector deployments.

Feasibility & Build Status

Data model and status-tracking workflow specified. Reference architecture is being built independently of Base44 on GitHub, deployable to any government IT environment without vendor lock-in — a deliberate design choice for public-sector procurement eligibility.

Grant & Funder Fit

Directly aligned to GovTech and anti-corruption funding lines: World Bank GovTech Global Partnership, Open Government Partnership grants, EU Digital Europe Programme, and USAID/DFID public-sector digitization funds.

3. TriageAI

SMS/USSD symptom triage, clinic routing, and blood-panel risk screening.

Key statistic: Millions of people die annually from cancers that are highly treatable when caught early, yet roughly half the world's population lacks access to basic diagnostic services (WHO).

Problem Statement

In low-connectivity and low-income regions, patients often have no accessible way to determine whether a symptom warrants urgent care, and existing blood test results are frequently filed without being reviewed for early risk markers due to clinician time pressure. TriageAI addresses both the front-door triage gap and the underused-data gap.

Technical Architecture

- SMS/USSD gateway enables symptom intake with zero smartphone or data requirement.
- Rule-based classifier maps reported symptoms to an urgency tier and routes to the nearest capable clinic.
- Premium blood-panel module compares patient-entered marker values against established clinical reference ranges and flags out-of-range results for specialist follow-up.

- All risk-flag outputs are explicitly labeled as a screening aid, never a diagnosis — by design, to remain outside full medical-device regulatory classification while still delivering real triage value.

Core Data Model

Entity	Purpose
Patient	Anonymized or consented patient profile and contact channel.
SymptomReport	Reported symptoms, urgency classification, timestamp.
ClinicDirectory	Verified clinic locations, capacity, contact details.
BloodPanelUpload	Patient-entered blood marker values (CBC, PSA, CEA, CA-125, AFP, etc.).
RiskFlag	Rule-based/ML-assisted flag against reference ranges, always paired with a specialist-referral disclaimer.

Security, Privacy & Compliance

- Health data encrypted at rest and in transit; access scoped strictly to the reporting patient and their nominated clinic.

- No blood-panel data used for anything beyond generating the immediate risk-flag output — no secondary use without explicit consent.
- Clear regulatory positioning as clinical decision support / screening aid, distinct from a diagnostic medical device, to align with a realistic and lawful path to market.

Feasibility & Build Status

Symptom-intake and clinic-routing logic specified; blood-panel reference-range comparison logic in design. Independent build on GitHub, with AI/NLP symptom classification model targeted for Hugging Face hosting.

Grant & Funder Fit

Strong fit for global health innovation funding: Gates Foundation Global Health Discovery, WHO digital health initiatives, EU Horizon Europe Health cluster, and national ministry of health digital triage programs.

4. LearnOffline

Offline-capable, curriculum-adaptive AI tutor for low-connectivity schools.

Key statistic: 617 million children and adolescents worldwide are not achieving minimum proficiency in reading or mathematics, many without ever having access to a qualified teacher (UNESCO).

Problem Statement

Rural and low-income schools frequently face teacher shortages and unreliable internet access, both of which compound learning loss. Existing ed-tech solutions largely assume constant connectivity, excluding exactly the students who need the most support.

Technical Architecture

- Lesson packs download once over any available connection, then operate with zero ongoing internet dependency.
- Adaptive-difficulty engine adjusts question pacing per student based on demonstrated mastery, functioning entirely on-device.
- Background sync protocol uploads progress automatically the instant connectivity is detected, with conflict-safe merge logic.

- Curriculum mapping layer allows content to be aligned to a selected national or regional syllabus.

Core Data Model

Entity	Purpose
Student	Learner profile, grade level, assigned curriculum.
Lesson	Curriculum-aligned content unit, downloaded for fully offline use.
ProgressRecord	Mastery score per lesson/topic, timestamped, queued for sync.
School	Institution profile, licensing tier, enrolled students.

Security, Privacy & Compliance

- Student data stored locally on-device until sync, minimizing exposure during offline periods.
- School-level access controls ensure teachers only see their own enrolled students' progress.
- No third-party advertising or data-resale — education data used solely for learning-outcome purposes.

Feasibility & Build Status

Offline content packaging and adaptive-difficulty logic specified. Build proceeding independently on GitHub; on-device inference model evaluation targeted for Hugging Face hosting and lightweight distillation for low-end hardware.

Grant & Funder Fit

Strong fit for education innovation funding: UNESCO/UNICEF EdTech innovation funds, EU Erasmus+ digital education calls, GPE (Global Partnership for Education), and national ministry of education digital learning grants.

5. SafeGuard

ScamShield fraud/romance-scam detection and SafeLink anonymous trafficking-victim support.

Key statistic: Romance and investment scams cost victims an estimated \$10+ billion annually worldwide, and an estimated 27.6 million people are in situations of forced labor or trafficking (ILO/Walk Free).

Problem Statement

Victims of romance and investment fraud typically have no real-time warning system before money is sent, and trafficking victims often lack a safe, anonymous first step to seek help. SafeGuard is deliberately built as a prevention and victim-support tool, not a private enforcement tool — narcotics-syndicate interdiction is explicitly out of scope, as that is a law-enforcement/intelligence function, not a viable or safe civilian software product.

Technical Architecture

- ScamShield: NLP pattern-matching engine scores incoming message text against known scam tactics (urgency framing, fake investment returns, gift-card/wire requests).
- ScamShield: perceptual image-hashing cross-references submitted profile photos against a

crowdsourced reported-scammer database, independent of filename or minor edits.

- SafeLink: anonymous, non-attributable tip submission channel, routed directly to NAPTIP, IOM, and the National Human Trafficking Hotline.
- SafeLink: location-based resource locator surfaces the nearest verified shelter or legal aid contact under time pressure.

Core Data Model

Entity	Purpose
ScamReport	Submitted message/profile pattern, similarity score against known tactics.
ReportedProfile	Crowdsourced record of a reported scammer photo/profile, cross-referenced across submissions.
TraffickingTip	Anonymized tip submission, routed to the appropriate hotline/agency.
ResourceLocator	Verified shelter, legal aid, and NGO support resource by location.

Security, Privacy & Compliance

- SafeLink tips are architecturally anonymous by default — no persistent identifying metadata is retained unless the reporter explicitly opts in to follow-up contact.
- ScamShield's reported-scammer database is moderated to prevent misuse for harassment; multiple corroborating reports required before a profile is surfaced.
- Explicit scope boundary: no surveillance, no narcotics-network targeting, no vigilante action — strictly prevention, detection, and routing to legitimate authorities.

Feasibility & Build Status

Pattern-detection and anonymous-reporting architecture specified. Independent build on GitHub; NLP scam-pattern classifier and image-hash matching model targeted for Hugging Face hosting.

Grant & Funder Fit

Strong fit for anti-trafficking and consumer-protection funding: US State Department Office to Monitor and Combat Trafficking in Persons (J/TIP) grants, EU Anti-Trafficking Action Grants, UNODC, and [Google.org/Meta](https://www.google.org/meta) digital safety grants — a distinct funder pool from the other four platforms.

6. ChipLink

Fabless mature-node chip sourcing and design brokerage.

Key statistic: A handful of companies control the world's advanced semiconductor supply. Leading-edge fabrication (below 7nm) requires ASML EUV lithography tooling, export-controlled to a small number of countries and costing \$10-20 billion+ per fab — but the mature-node tier (28nm and above), which powers most real-world devices including payment terminals, automotive systems, and IoT, remains open to new entrants.

Problem Statement

Small and mid-sized hardware manufacturers, including African fintech and IoT makers, are frequently locked out of chip supply by minimum-order-quantity thresholds and broker markups, even for mature-node components that do not require cutting-edge fabrication. ChipLink deliberately does not attempt to compete at the leading edge, which is not a realistic or safe target for a company at this stage — instead it targets the large, genuinely accessible mature-node and chip-packaging/testing (OSAT) market.

Technical Architecture

- RFQ marketplace lets a buyer submit a chip specification and quantity; verified mature-node foundries and OSAT partners bid competitively and transparently.
- Order-pooling engine aggregates volume across multiple small buyers with compatible specifications to collectively reach a foundry's minimum-order-quantity threshold.
- Fabless design-support layer helps a buyer without in-house chip expertise translate a product requirement into a manufacturable specification.
- Explicit scope boundary: ChipLink targets mature-node (28nm and above) sourcing and OSAT packaging/testing only — it does not attempt leading-edge fabrication, which requires export-controlled tooling outside realistic reach.

Core Data Model

Entity	Purpose
Buyer	Hardware manufacturer submitting a chip specification and required volume.
ChipSpec	Technical specification: process node, package type, functional requirements.

FoundryPartner	Verified mature-node foundry or OSAT partner, capacity and lead time.
RFQ	Request-for-quote record with competing bids from foundry partners.
OrderPool	Aggregated volume from multiple buyers pooled to reach a foundry's minimum order quantity.

Security, Privacy & Compliance

- Buyer chip specifications treated as confidential; visible only to bidding foundry partners under a standard non-disclosure framework.
- Foundry partner verification process to guard against counterfeit or non-compliant component sourcing.
- Order-pooling logic ensures no single pooled buyer can see another pooled buyer's full specification or pricing terms.

Feasibility & Build Status

RFQ marketplace and order-pooling logic specified. Independent build on GitHub; buyer-foundry matching/ranking logic targeted for Hugging Face hosting. This platform also directly serves

Zeropay/BETONIQ WEST's own hardware supply chain need for PUF/ECC secure element components.

Grant & Funder Fit

Fits industrial policy and supply-chain resilience funding: US CHIPS Act adjacent programs, EU Chips Act supply-chain diversification funds, India Semicon Mission, and African Union/AfDB regional value-chain development grants — a distinct funder pool from the other five platforms.

Group Summary & Next Steps

All six platforms are structured as independent product lines under Zeropay Digital Services Ltd, the UK subsidiary of BETONIQ WEST LTD. This structure allows each platform to be pitched to its own most relevant funder category — agriculture, GovTech, global health, education, anti-trafficking/consumer protection, and semiconductor supply-chain resilience — while sharing common engineering standards: GitHub-based version control, Hugging Face-hosted AI inference, offline-first design where relevant, and a consistent commitment to data minimization and privacy by design.

Immediate next steps: finalize Hugging Face deployment access; complete first working prototypes in build order (AgriLink, CivicTrack, TriageAI, LearnOffline, SafeGuard, ChipLink); and submit the first wave of grant applications referencing this white paper as supporting technical documentation.